

JSH KAPIL AGARWAL

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Education

Sardar Patel Institute of Technology (7.88 CGPA) <i>Bachelor of Technology in Electronics and Telecommunication</i>	12/2022 – 06/2026 <i>Mumbai, Maharashtra</i>
SP Jain Institute of Management <i>Minors in Management</i>	01/2024 – 01/2026 <i>Mumbai, Maharashtra</i>

Experience

SmowCode <i>Software Developer Intern</i>	06/2024 – 11/2024 <i>Remote</i>
- Developed scalable PostgreSQL + Flask backend for LMS portal serving 500+ users, implementing secure course management and automated certificate generation processing 100+ certificates monthly. - Built Scarlet, a Generative AI feature utilizing RAG architecture and ChromaDB, achieving 95% user satisfaction through effective stakeholder presentations and user feedback integration across 200 queries. - Created Esperin, an AI code assistant for embedded developers using Agentic workflows; delivered technical documentation to cross-functional teams and optimized LLMs with LoRA & QLoRA, reducing debugging time by 40%.	

Projects

AlphaSage <i>Flask, MongoDB, ChromaDB, AutoGen, Mistral OCR, LM Studio</i>	05/2025
- Architected multi-agent AI platform automating equity research for Indian financial markets using Groq API and yfinance integration,eliminating 70% of manual analysis workflows and saving 40+ hours weekly. - Designed unified JSON schema parsing system for textual, tabular, and visual data extraction from investor filings, achieving 90% accuracy across 200+ document samples and processing 50+ filings daily. Demonstrated findings to industry panel at JPMC Tech Week 2025 among Top 4 projects across Mumbai region, communicating complex financial analytics insights to 15+ technical and non-technical stakeholders.	
RakshaSetu <i>Python, Flask, Gemini 2.0 Flash, Scikit-learn, Folium,Cryptography (Fernet)</i>	04/2025
- Developed secure Flask backend with RESTful API architecture for real-time crime reporting system serving 1,000+ users, integrating Fernet-based encryption protocols for and reducing crime response time by 35%. - Integrated Gemini 2.0 Flash for automated crime classification across 15+ crime categories and summary generation from multimedia inputs, improving detection accuracy by 25% and processing 200+ reports daily. - Deployed geospatial visualization dashboard using Folium framework, enabling law enforcement agencies to analyze crime patterns across 50+ locations with 85% location precision, resulting in 15% faster case resolution.	
Warehouse Management System <i>Python, OpenCV, YOLOv8, ConvLSTM, Keras</i>	01/2025
- Implemented ensemble model combining YOLOv8 + 2D ConvLSTM for object detection and quantity estimation across 500+ product categories, achieving 92% detection accuracy and reducing inventory discrepancies by 60%. - Built comprehensive dashboard with time-series analytics powered by BiLSTM networks for inventory monitoring of 1,000+ SKUs, decreasing stockout incidents by 45% with real-time updates	

Awards and Achievements

Winner – Odoo X Mindbend, SVNIT Surat: Recognized for **RakshaSetu** project, securing **top 3 position** among **600+ participating teams** in prestigious **all-India hackathon** competition .

Winner - Tecnovanza: Achieved **2nd place** in elite competition evaluated by **industry experts**,delivered compelling **15-minute presentation** to panel of **15+ judges**, securing **50,000 prize**.

Winner - JamRang 2025: Ranked among **top 3 teams** from **450+ participants** in national **GenAI hackathon** hosted by **VijayBhoomi University**, articulated technical innovation through **live demo and Q&A session** to diverse stakeholder audience.

Positions of Responsibility

Finance and Economics Club, SPIT <i>Vice President – Tech</i>	09/2024 - Present
- Spearheaded development of 5 machine learning models for stock market analysis covering NSE top 100 stocks; authored and presented comprehensive research reports to 200+ club members and faculty stakeholders. - Mentored team of 8 junior developers, conducting weekly training sessions and coordinating website development using modern frameworks, accelerating development time by 30% through technical guidance.	